

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276837

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Oefinger; Terence

System and Method for Containing Multiple Fluids

Abstract

A replacement lid and insert system for an existing fluid container includes a container lid with a straw compartment, a pour compartment, and a threaded portion compatible with the existing fluid container. A straw extends from the lid into the fluid container. A reservoir is in fluid communication with the pour compartment, the reservoir having an exterior shaped to reside within the fluid container and an interior segregated from the container interior. The reservoir can be coupled to the container lid through a reservoir interface or can be integral with the lid. The system may also include a container seal that creates a fluid-tight seal when the lid is coupled to the fluid container, a pour lid selectively covering the pour compartment, and a straw mouthpiece fluidly coupled to the straw.

Inventors: Oefinger; Terence (South Jordan, UT)

Applicant: Oefinger; Terence (South Jordan, UT)

Family ID: 96880909

Appl. No.: 18/592765

Filed: March 01, 2024

Publication Classification

Int. Cl.: B65D51/28 (20060101); B65D47/30 (20060101); B65D51/24 (20060101); B65D77/06 (20060101); B65D85/72 (20060101)

U.S. Cl.:

CPC B65D51/28 (20130101); B65D47/30 (20130101); B65D51/242 (20130101); B65D77/06 (20130101); B65D85/72 (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention generally relates to a replacement lid and insert system for fluid containers, and more specifically to a system that allows for the segregation of two different fluids within the same container.

Description of Related Art

[0002] Containers may hold a variety of different types of liquids such as water, beverages, drinks, juices, and the like. Containers also may hold various items such as energy drinks, protein drinks, shakes, foodstuffs, dressings, sauces, and liquid meal replacement. Known containers may be used in a wide variety of environments such as at a home, office, gym, or health club, and while traveling. Known beverage containers may also be used during or after activities such as exercising or riding in an automobile, bus, train, or airplane.

[0003] A lid with a closure may be used to control access to an interior of the container. The lid may be removed entirely to fill the container with ice or other contents, to wash the container, or to otherwise provide access to the interior of the container through the relatively large opening of the container. The closure may be opened to allow a user to consume contents of the container through the relatively smaller opening of the lid or to otherwise provide access to the interior of the container through the relatively smaller opening formed in the lid.

[0004] Some known beverage containers may include a small cover over the drinking aperture to prevent the contents from spilling. The small cover is typically manually opened by the user's hand when the user desires to take a drink. For example, the user may slide the cover into an open position. Many of these known containers include a watertight or airtight seal between the cover and the drinking aperture to prevent the beverage from spilling. These known containers often require substantial effort to open and close the cover because the watertight or airtight seal must be formed or broken each time the cover is opened or closed. Some of these containers, especially when the container is full, may undesirably allow some of the beverage to spill when the cover is opened.

[0005] In addition, because the cover may be suddenly and abruptly opened, this may cause some of the contents to spill. Further, these known beverage containers may be difficult for some consumers to use because a relatively large amount of force may be required to open or close the cover. Thus, some consumers may have substantial difficulty opening the cover. Additionally, the cover of some of these known containers may be difficult to securely close and that unfortunately may allow the contents to spill or leak.

[0006] Known beverage containers may also use a cover that rotates between an open and closed position. Disadvantageously, the cover may be difficult to open, close and/or rotate into the desired positions. Conventional beverage containers may also include a button or lever that must be depressed to open the cover and allow the beverage to be consumed. Some beverage containers undesirably require the button to be continuously depressed in order to drink from the container, which is often awkward and uncomfortable for the user.

[0007] Conventional beverage containers may also include rotating spouts and removable lids, but these types of devices are often difficult to open or close, especially for users with limited dexterity. In particular, it may be challenging or almost impossible for some users to rotate the spout from a closed position to an open position because of the difficulty in grasping the spout. It may also be problematic for many users to apply the force required to move the spout into the open position. Further, a user may not want to touch the spout with his or her fingers if their hands are dirty or unwashed. It may also be difficult for some users to remove the lid to allow access to the contents of the container. Further, it may be tedious to constantly remove the lid before drinking and this may prevent liquids from being consumed while exercising, traveling, during sporting events, or when a quick or rapid drink is desired.

[0008] Further, conventional beverage containers may have limited usefulness because they may only be used for specific purposes. Some conventional beverage containers may also be difficult to clean or reuse because of the shape, number of parts, and complex configuration of one or more components. In some instances, conventional containers may hold fluids that require mixing. For example, conventional containers may hold powdered drinks, electrolyte pills, energy drinks, baby formulas, pancake batters, crepe mixes, baking materials, dietary supplements, salsa, sauces, oil and vinegar, salad dressings, smoothies, and the like. These types of containers that hold fluids for mixing may require the lid to be securely closed and the lid to be fluid-tight so that fluid does not leak from the lid.

[0009] Some known containers that are designed to mix ingredients by shaking the container may include lids that are difficult and time consuming to open and close. For example, the lid may be screwed on the container and the user may have to unscrew the lid to dispense the contents, which may be especially difficult for users with limited agility or dexterity. Some known containers may not be leak-proof and the contents may leak when shaking the container or if the container is placed at an angle.

[0010] What is needed is a simple inexpensive multiple chamber container with separate fluid chambers selectively accessible by a user and a removable cap sealing the two chambers separately and sealable fluid access openings in the cap to access fluid individually from each of the separate fluid chambers while maintaining the other chamber sealed.

[0011] So as to reduce the complexity and length of the Detailed Specification, and to fully establish the state of the art in certain areas of technology, Applicant(s) herein expressly incorporate(s) by reference all of the following materials identified in each numbered paragraph below. The incorporated materials are not necessarily “prior art” and Applicant(s) expressly reserve(s) the right to swear behind any of the incorporated materials.

[0012] U.S. Patent Application #20060175350, published Aug. 10, 2006 by Abramson

[0013] U.S. Pat. No. 6,820,767, issued Nov. 23, 2004 to Nicholas

[0014] U.S. Pat. No. 5,758,786, issued Jun. 2, 1998 to John

[0015] U.S. Pat. No. 5,890,627, issued Apr. 6, 1999 to Storey

[0016] U.S. Pat. No. 6,079,586, issued Jun. 27, 2000 to Hanneman

[0017] Two U.S. patents, U.S. Pat. No. 5,753,289 issued May 19, 1998 and U.S. Pat. No. 5,588,561 issued Dec. 31, 1996 to Ness,

[0018] U.S. Pat. No. 5,593,052, issued Jan. 14, 1997 to McGee

[0019] U.S. Pat. No. 7,090,097 issued Aug. 15, 2006 and U.S. Pat. No. 6,769,573 issued Aug. 3, 2004 to Kazarian

[0020] U.S. Patent Application #20060131311, published Jun. 22, 2006 by Wang,

[0021] U.S. Patent Application #20060065660, published Mar. 30, 2006 by Kalaouze

[0022] U.S. Patent Application #20070029275, published Feb. 8, 2007 by Hantman

[0023] U.S. Patent Application #20020074347, published Jun. 20, 2002 by Murray

[0024] U.S. Pat. No. 6,863,193, issued Mar. 8, 2005 to Grossenbacher

[0025] U.S. Pat. No. 6,913,777, issued Jul. 5, 2005 to Rebhorn

[0026] U.S. Pat. No. 7,036,684, issued May 2, 2006 to Hantman

[0027] U.S. Pat. No. 6,152,862 issued Nov. 28, 2000 and U.S. Pat. No. 7,172,533 issued Feb. 6, 2007 to Hobbs

[0028] U.S. Pat. No. 4,301,942, issued Nov. 24, 1981 to Kupperman

[0029] U.S. Pub. No. 2013/0001233 filed Nov. 5, 2012 to Hylton,

[0030] Applicant(s) believe(s) that the material incorporated above is “non-essential” in accordance with 37 CFR 1.57, because it is referred to for purposes of indicating the background of the invention or illustrating the state of the art. However, if the Examiner believes that any of the above-incorporated material constitutes “essential material” within the meaning of 37 CFR 1.57(c) (1)-(3), applicant(s) will amend the specification to expressly recite the essential material that is

incorporated by reference as allowed by the applicable rules.

BRIEF SUMMARY OF THE INVENTION

[0031] In accordance with one embodiment, the invention provides a replacement lid and insert system for use with an existing fluid container. The system includes a container lid with a straw compartment, a pour compartment, and a threaded portion designed to be compatible with the existing fluid container. A straw extends from the lid into the fluid container. The system also includes a reservoir that is in fluid communication with the pour compartment. The reservoir has an interior and an exterior, with the exterior shaped to reside within the fluid container. The reservoir interior is segregated from the container interior. In some embodiments, the reservoir is coupled to the container lid through a reservoir interface. The reservoir interface may comprise an internally threaded extension that corresponds to an externally threaded extension on the reservoir. In other embodiments, the reservoir is integral with the lid. The system may further include a container seal that creates a fluid-tight seal when the lid is coupled to the fluid container.

[0032] The system may also include a pour lid that selectively covers the pour compartment and a straw mouthpiece fluidly coupled to the straw, allowing selective access to the straw. The reservoir may have a rigid fixed volume or be a flexible bladder. In some embodiments, the reservoir has the same shape and volume of a traditional hip flask. In other embodiments, the reservoir is configured to have the same volume in the reservoir interior as the volume of the container interior that is not filled with the insert system.

[0033] The straw mouthpiece may be hingedly coupled to a handle that allows a user to pull the straw mouthpiece from a closed position to an open position. The pour lid and/or the straw mouthpiece may maintain an elevated pressure to allow a carbonated beverage to be stored in the container interior and/or the reservoir.

[0034] The replacement lid and insert system for use with an existing fluid container having a container interior also provides a method of segregating at least two liquids in an existing fluid container. The method includes providing the replacement lid and insert system, filling the reservoir with a first fluid, partially filling the fluid container interior with a second fluid, inserting the reservoir into the container interior, and securing the container lid to the fluid container. In some embodiments, the first fluid is an alcoholic beverage. The method may further include providing a carbonated fluid into the container interior or the reservoir and selectively sealing the container interior or the reservoir containing the carbonated fluid to maintain an elevated pressure.

[0035] Aspects and applications of the invention presented here are described below in the drawings and detailed description of the invention. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts. The inventors are fully aware that they can be their own lexicographers if desired. The inventors expressly elect, as their own lexicographers, to use only the plain and ordinary meaning of terms in the specification and claims unless they clearly state otherwise and then further, expressly set forth the “special” definition of that term and explain how it differs from the plain and ordinary meaning. Absent such clear statements of intent to apply a “special” definition, it is the inventors' intent and desire that the simple, plain and ordinary meaning to the terms be applied to the interpretation of the specification and claims.

[0036] The inventors are also aware of the normal precepts of English grammar. Thus, if a noun, term, or phrase is intended to be further characterized, specified, or narrowed in some way, then such noun, term, or phrase will expressly include additional adjectives, descriptive terms, or other modifiers in accordance with the normal precepts of English grammar. Absent the use of such adjectives, descriptive terms, or modifiers, it is the intent that such nouns, terms, or phrases be given their plain, and ordinary English meaning to those skilled in the applicable arts as set forth above.

[0037] Further, the inventors are fully informed of the standards and application of the special

provisions of 35 U.S.C. § 112 (f). Thus, the use of the words “function,” “means” or “step” in the Detailed Description or Description of the Drawings or claims is not intended to somehow indicate a desire to invoke the special provisions of 35 U.S.C. § 112 (f), to define the invention. To the contrary, if the provisions of 35 U.S.C. § 112 (f) are sought to be invoked to define the inventions, the claims will specifically and expressly state the exact phrases “means for” or “step for, and will also recite the word “function” (i.e., will state “means for performing the function of . . . ”), without also reciting in such phrases any structure, material or act in support of the function. Thus, even when the claims recite a “means for performing the function of . . . ” or “step for performing the function of . . . ,” if the claims also recite any structure, material or acts in support of that means or step, or that perform the recited function, then it is the clear intention of the inventors not to invoke the provisions of 35 U.S.C. § 112 (f). Moreover, even if the provisions of 35 U.S.C. § 112 (f) are invoked to define the claimed inventions, it is intended that the inventions not be limited only to the specific structure, material or acts that are described in the preferred embodiments, but in addition, include any and all structures, materials or acts that perform the claimed function as described in alternative embodiments or forms of the invention, or that are well known present or later-developed, equivalent structures, material or acts for performing the claimed function.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0038] A more complete understanding of the present invention may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the figures, like reference numbers refer to like elements or acts throughout the figures.

[0039] FIG. 1 depicts an exploded perspective view of an insert according to one embodiment of the invention.

[0040] FIG. 2 depicts an exploded perspective view of another embodiment of an insert according to one embodiment of the invention.

[0041] FIG. 3 depicts an overhead perspective view of an insert according to one embodiment of the invention;

[0042] FIG. 4 depicts a side view of an insert according to one embodiment of the invention;

[0043] FIG. 5a depicts a front view of an insert according to one embodiment of the invention;

[0044] FIG. 5b depicts a cutaway view of FIG. 5a of an insert according to one embodiment of the invention;

[0045] FIG. 6 depicts an exploded side view of an insert according to one embodiment of the invention;

[0046] FIG. 7 depicts a side view of an insert in its closed, shot mode, straw mode position according to one embodiment of the invention;

[0047] FIG. 8 depicts an isometric view and side view of an insert according to one embodiment of the invention;

[0048] FIG. 9 depicts an isometric view of different embodiments of an insert according to one embodiment of the invention; and

[0049] FIG. 10 depicts an isometric view of various embodiments according to one embodiment of the invention.

[0050] Elements and acts in the figures are illustrated for simplicity and have not necessarily been rendered according to any particular sequence or embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0051] In the following description, and for the purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the various aspects of the invention. It will be understood, however, by those skilled in the relevant arts, that the present invention may be

practiced without these specific details. In other instances, known structures and devices are shown or discussed more generally to avoid obscuring the invention. In many cases, a description of the operation is sufficient to enable one to implement the various forms of the invention, particularly when the operation is to be implemented in software. It should be noted that there are many different and alternative configurations, devices, and technologies to which the disclosed inventions may be applied. The full scope of the inventions is not limited to the examples that are described below.

[0052] Referring to FIGS. **1-6**, an insert is shown generally at **10**. In one application of the invention, the insert **10** can be used with an existing fluid container. The insert **10** includes a main lid **12** that includes a threaded portion **13** configured to be compatible with common existing fluid containers. The main lid **12** has a straw compartment **15** and a pour compartment **17**. The straw compartment **15** houses a straw interface **14** that accommodates a straw **16** that extends from the main lid **12** into the container. The straw **16** may exceed common straw lengths and be cut to any appropriate length to fit a particular container. In certain embodiment the straw **16** can be omitted or can be varying diameters.

[0053] A straw mouthpiece **18** allows fluid access from the straw interface **14** by a user. In a particular embodiment, the straw mouthpiece **18** allows selective fluid access to the straw **16** via a valve **19** in the straw mouthpiece **18** that allows fluid access to the straw **16** when the straw mouthpiece **18** is in an open position and prevents fluid access to the straw **16** when the straw mouthpiece **18** is in a closed position.

[0054] The pour compartment **17** may be covered by a pour lid **20**. The pour lid **20** may be a hinged snap-shut lid, nib, or other suitable lid known to those having skill in the art. The pour compartment **17** is in fluid communication with a reservoir interface **22** which is in turn in fluid communication with a reservoir access **24**. The reservoir access **24** is in fluid communication with a reservoir **26**. The reservoir **26** has an interior and an exterior, with the exterior shaped to reside within the fluid container. The reservoir interior is segregated from the container interior.

[0055] The reservoir interface **22** may be made of the same material as the main lid **12** and be ultrasonically welded to the main lid **12**. The reservoir interface **22** may be a generally circular extension that is internally threaded to match a corresponding external threading on the reservoir access **24**. The reservoir interface **22** can have a reservoir seal **33** between the reservoir **26** and the reservoir interface. The reservoir seal **33** can stop liquids from entering or exiting the reservoir **26**. The reservoir **26** together with the straw **16** may have a circumference that is smaller than the inner circumference of the existing fluid container such that the straw **16** and the reservoir **26** are housed within the existing fluid container when the main lid **12** is coupled to the existing fluid container at the opening of the existing fluid container. Alternatively, the reservoir **26** may be integral to the main lid **12** and sit above the opening of the existing container when the main lid **26** is coupled to the existing fluid container as shown in FIG. **7**.

[0056] The insert **10** may include a container seal **27** that creates a fluid-tight seal between the main lid **12** and the container, a pour lid seal **28** that creates a fluid tight seal between the pour lid **20** and the pour compartment **29**, and a reservoir seal **33** that creates a fluid-tight seal between the reservoir access **24** and the reservoir interface **22**. The container seal **27** and the reservoir seal **22** can be made from such as, for example, rubber, Buna-N, neoprene, polyurethane, Teflon, or the like.

[0057] The reservoir **26** is shaped to conveniently move in and out of a common existing fluid container. The reservoir **26** may have a rigid shape that is sized to fit within the common fluid container, or it may be a flexible bladder or have flexible portions. In a particular embodiment, the reservoir **26** has the same volume as a traditional hip flask. The reservoir **26** may be configured to have a volume equal to the volume of the container that is not occupied by the insert **10**.

[0058] The insert **10** allows a user to carry two separate fluids in a single existing common container. In a particular embodiment, the straw mouthpiece **18** is hingedly coupled to a handle **30**

that allows a user to pull the straw mouthpiece **18** more easily from a closed position to an open position and in other embodiments the straw mouthpiece can be such as, for example, slideably, removable, or the like to the handle or to the main lid. A connection hook **32** is coupled by, for example, ultrasonic welding, to the main lid **12**. The connection hook **32** is positioned such that the handle **30** may form a loop with the connection hook **32**.

[0059] In use, a user may couple the insert **10** with a common fluid container to effectively house two separate fluids. For example, the reservoir **26** may contain some Seagram's gin, and the common container may contain fruit juice. The common container may be prefilled with fluid or may be filled after the insert is in place. The straw mouthpiece **18** and/or the pour lid **20** may be configured to maintain an elevated pressure to allow carbonated beverages to be stored in the container and/or the reservoir **26**. A vent cap **36** may provide a vent between the open air and the reservoir interface **24**.

[0060] The user may fill the reservoir **26** with a first fluid and couple the reservoir **26** to the reservoir interface **24**. The user may also fill the common container at partially with a second fluid. The insert **10** may be inserted into the common container and the main lid **12** secured to the common fluid container. The user may selectively access the second fluid through the straw mouthpiece **18** and the first fluid through the pour compartment **17**.

[0061] As shown in FIG. **7** the insert can allow the user to access fluids through the straw mouthpiece **18** when in straw mode, can allow the user access to another fluid within the reservoir when in shot mode when the pour compartment **17** is open or can have both closed keeping the liquids within the reservoirs. FIGS. **8-10** show other embodiments of the insert **10** in its closed, open to a fluid, or open to the fluid in the reservoir.

[0062] The insert also comprises a method of segregating at least two liquids in an existing fluid container. The method includes providing the replacement lid **12** and insert system, filling the reservoir with a first fluid, partially filling the fluid container interior with a second fluid, inserting the reservoir into the container interior, and securing the container lid **12** to the fluid container. In some embodiments, the first fluid can be an alcoholic beverage. The method may further include providing a carbonated fluid into the container interior or the reservoir and selectively sealing the container interior or the reservoir containing the carbonated fluid to maintain an elevated pressure.

[0063] In closing, it is to be understood that although aspects of the present specification are highlighted by referring to specific embodiments, one skilled in the art will readily appreciate that these disclosed embodiments are only illustrative of the principles of the subject matter disclosed herein. For example, alternative configurations of the main lid **12** are shown in FIGS. **7-10**. As another example, the two separate fluids may be mixed as they are being sucked out of the container. Therefore, it should be understood that the disclosed subject matter is in no way limited to a particular methodology, protocol, and/or reagent, etc., described herein. As such, various modifications or changes to or alternative configurations of the disclosed subject matter can be made in accordance with the teachings herein without departing from the spirit of the present specification. Lastly, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the scope of the present disclosure, which is defined solely by the claims. Accordingly, embodiments of the present disclosure are not limited to those precisely as shown and described.

[0064] Certain embodiments are described herein, including the best mode known to the inventors for carrying out the methods and devices described herein. Of course, variations on these described embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. The terms "including" and "such as" are not limiting and should be interpreted as "including, but not limited to," and "such as, for example," respectively. Moreover, any combination of the above-described

embodiments in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A replacement lid and insert system for use with an existing fluid container having a container interior, the system comprising: a container lid having a straw compartment, a pour compartment, and a threaded portion configured to be compatible with the existing fluid container; a straw extending from the lid into the existing fluid container; a reservoir in fluid communication with the pour compartment and having a reservoir interior and a reservoir exterior, wherein the reservoir exterior is shaped to reside within the fluid container and wherein the reservoir interior is segregated from the container interior.
 2. The replacement lid and insert system of claim 1, wherein the reservoir is coupled to the container lid through a reservoir interface.
 3. The replacement lid and insert system of claim 2, wherein the reservoir interface comprises an internally threaded extension that corresponds to an externally threaded extension on the reservoir.
 4. The replacement lid and insert system of claim 1, wherein the reservoir is integral with the lid.
 5. The replacement lid and insert system of claim 1, further comprising a container seal that creates a fluid-tight seal when the lid is coupled to the fluid container.
 6. The replacement lid and insert system of claim 1, further comprising a pour lid selectively covering the pour compartment and a straw mouthpiece fluidly coupled to the straw and that allows selective access to the straw.
 7. The replacement lid and insert system of claim 1, wherein the reservoir has a rigid fixed volume or is a flexible bladder.
 8. The replacement lid and insert system of claim 1, wherein the reservoir has the same shape and volume of a traditional hip flask.
 9. The replacement lid and insert system of claim 1, wherein the reservoir configured to have the same volume in the reservoir interior as the volume of the container interior that is not filled with the insert system.
 10. The replacement lid and insert system of claim 6, wherein the straw mouthpiece is hingedly coupled to a handle that allows a user to pull the straw mouthpiece from a closed position to an open position.
 11. The replacement lid and insert system of claim 6, wherein the pour lid and/or the straw mouthpiece maintains an elevated pressure to allow a carbonated beverage to be stored in the container interior and/or the reservoir.
 12. A method of segregating at least two liquids in an existing fluid container having a container interior, the method comprising the acts of: providing a replacement lid and insert system of claim 1; filling the reservoir with a first fluid; partially filling the fluid container interior with a second fluid; inserting the reservoir into the container interior and securing the container lid to the fluid container.
 13. The method of claim 12, wherein the first fluid is an alcoholic beverage.
 14. The method of claim 12 wherein the method further comprises: providing the replacement lid and insert system with a pour lid selectively covering the pour compartment and a straw mouthpiece fluidly coupled to the straw and that allows selective access to the straw; providing a carbonated fluid into the container interior or the reservoir; and selectively sealing the container interior or the reservoir containing the carbonated fluid to maintain an elevated pressure.
-